Filtering the 2023-2024 dataset for Aphis fabae

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Summary

Calculation of key metrics based on the 2023-2024 Aphis fabae phenotyping of the PEAMust collection.

1) Calculation of the broad-sense heritability ("repeatability") as a function of the number of biological replicates.
2) Calculation of the correlation between sessions.
3) Decision to keep N biological replicates: compromise between repeatability and number of genotypes/observations.

Setup

Packages

This analysis depends on the following packages:

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("patchwork"))
```

Helper functions

Two helper functions are created:
- `calculate_broad_heritability()` giving the broad sense heritability (H^2) and the number of genotypes of a given dataset.
- `create_correlation_plot()` to calculate the correlations between experimental sessions.

Broad-sense heritability calculations

Raw dataset

```
data250pisum_00 <- read.csv("../../../../../01.data/2023_screen_specifics_251genotypes/02.apl
  mutate(genotype = as.factor(genotype)) %>%
  mutate(session = as.factor(session))

res <- calculate_broad_heritability(data = data250pisum_00)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.097.
> - There are 222 genotypes in total.
> - There are 2664 measurements in total.

Missing data

Remove NA values.

```
data250pisum_01 <- data250pisum_00 %>% filter(!is.na(larvae_total_number))
data250pisum_01$genotype <- droplevels(data250pisum_01$genotype)

res2 <- calculate_broad_heritability(data = data250pisum_01)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.097.
> - There are 220 genotypes in total.
> - There are 2370 measurements in total.

```
create_correlation_plot(data250pisum_01,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

Remove observations with no adult alive

We chose to remove all data from plants on which no adult aphid was found. This decision was made to get rid of larvae installation bias and only assess the reproduction ability of the aphid.

When the adult aphid dies, the number of larvae will drop but only because the adult stayed less longer on the plant and not because the plant was not suitable for the reproduction.

```
data250pisum_02<- data250pisum_01 %>% filter(adult_alive != "0") #remove if no adult
data250pisum_02$genotype <- droplevels(data250pisum_02$genotype)

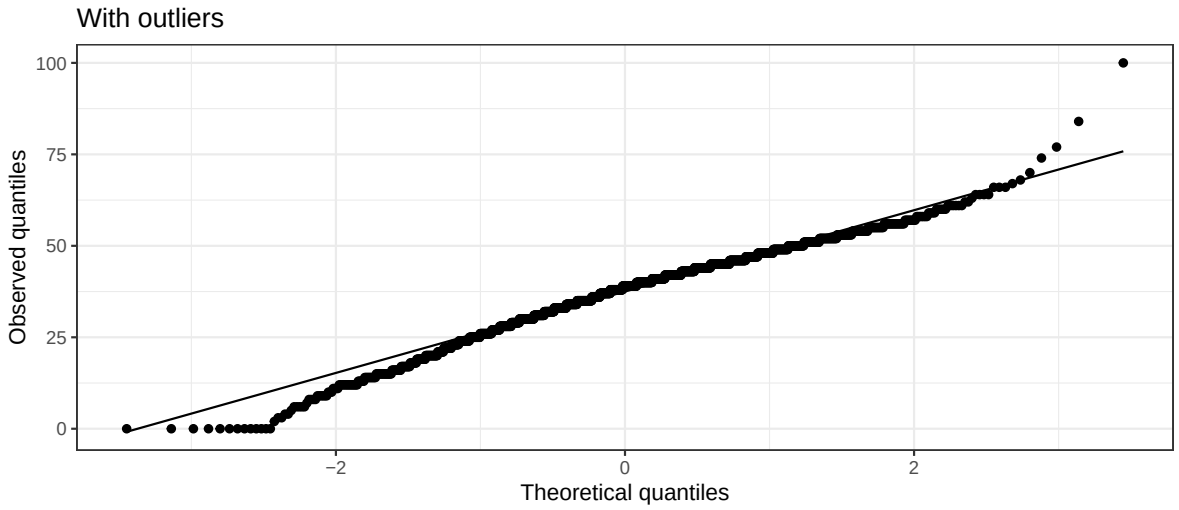
res3 <- calculate_broad_heritability(data = data250pisum_02)
```

> Note
>
> Conclusions:
> - Heritability is equal to 0.289.
> - There are 220 genotypes in total.
> - There are 1768 measurements in total.

Heritability strongly increases. 602 measurements are lost. The number of genotype stays the same.

```
create_correlation_plot(data250pisum_02,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

```
# With outliers
p_with <- data250pisum_02 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("With outliers") +
  theme_bw()
p_with
```



A few data outliers (n=3) with high larvae number values seem to be problematic. Let's remove the top 3.

```
# Identify the indices of the top 3 highest values
top_3_indices <- order(data250pisum_02$larvae_total_number, decreasing = TRUE)[1:3]

# View the top 3 highest values
data250pisum_02[top_3_indices, ] %>% dplyr::select(genotype, ivip, aphid, larvae_total_numbe
```

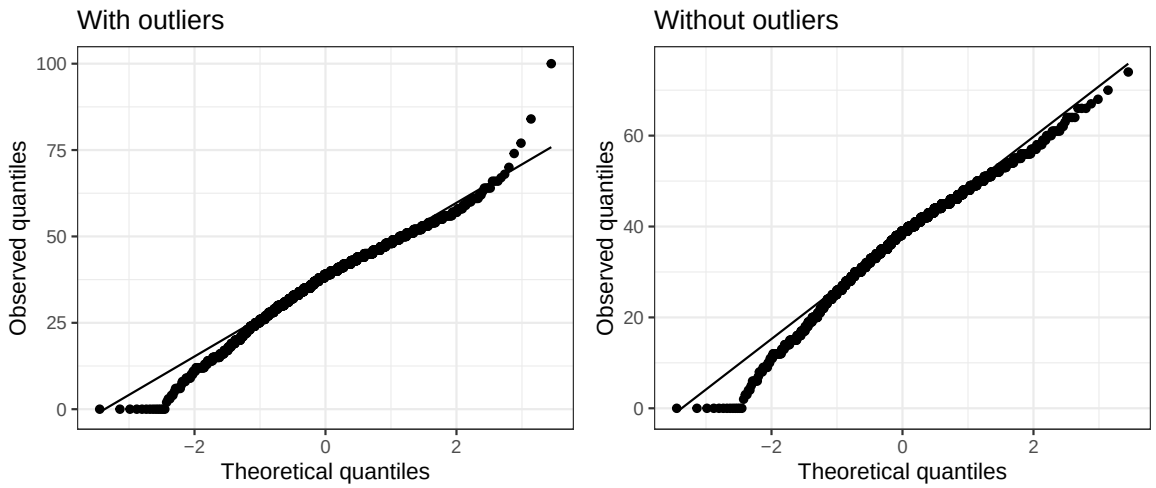
```
     genotype       ivip         aphid larvae_total_number
```

```
1146      164 IVIP101838 A.fabae AF001                 100
1089      155 IVIP101738 A.fabae AF001                  84
123        19 IVIP100234 A.fabae AF001                  77
```

```
# Remove the rows with the top 3 highest values
data250pisum_03 <- data250pisum_02[-top_3_indices, ]
```

```
# Without outliers
p_without <- data250pisum_03 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Without outliers") +
  theme_bw()

p_with + p_without
```



Finding the best number of biological replicates criteria

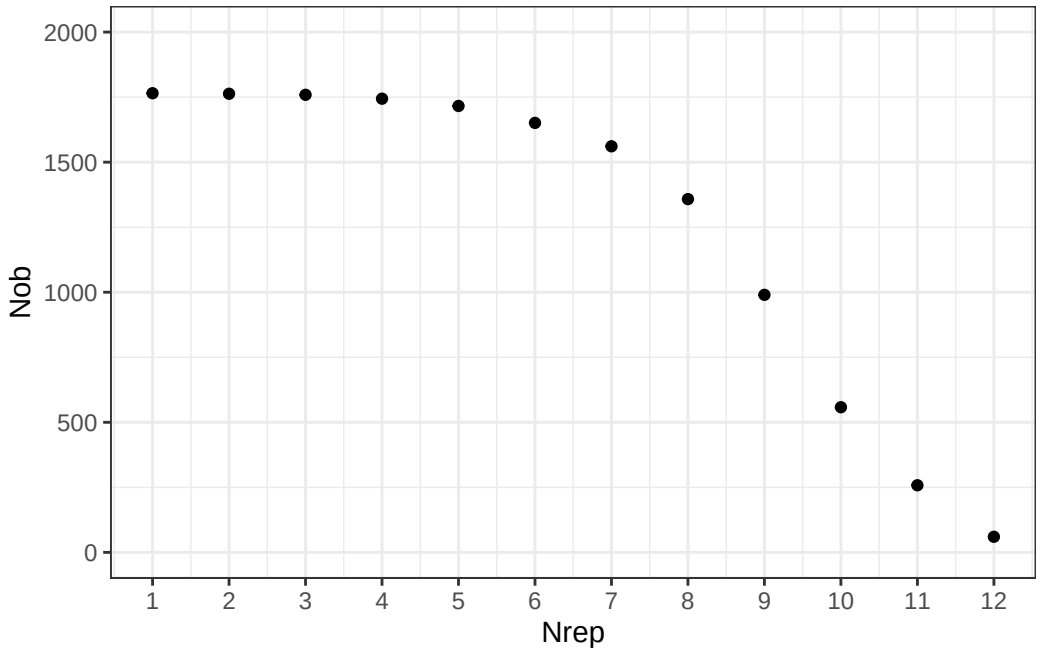
Now the question is: should we filter genotypes for which we have less than X biological repetitions?

Let's extract:
- The number of genotypes available for X number of biological replicates.

- The number of observations available for X number of biological replicates.
- The broad-sense heritability for X number of biological replicates.

N observations = f(N biological replicates)

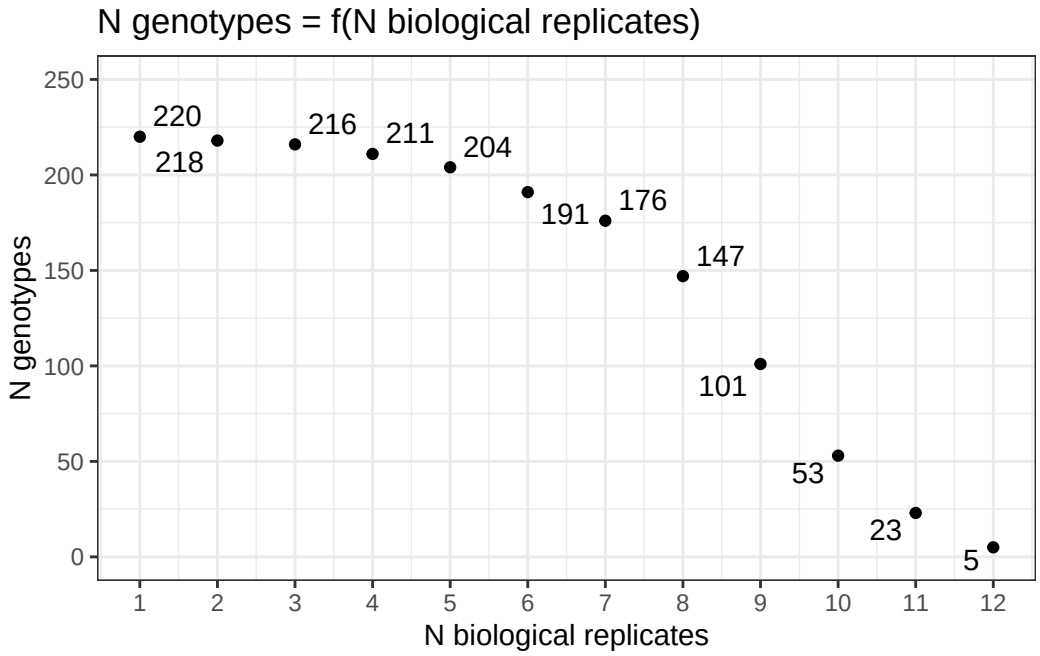
```
ggplot(dfb, aes(x = Nrep, y = Nob)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,2000)) +
  theme_bw()
```



N genotypes = f(N biological replicates)

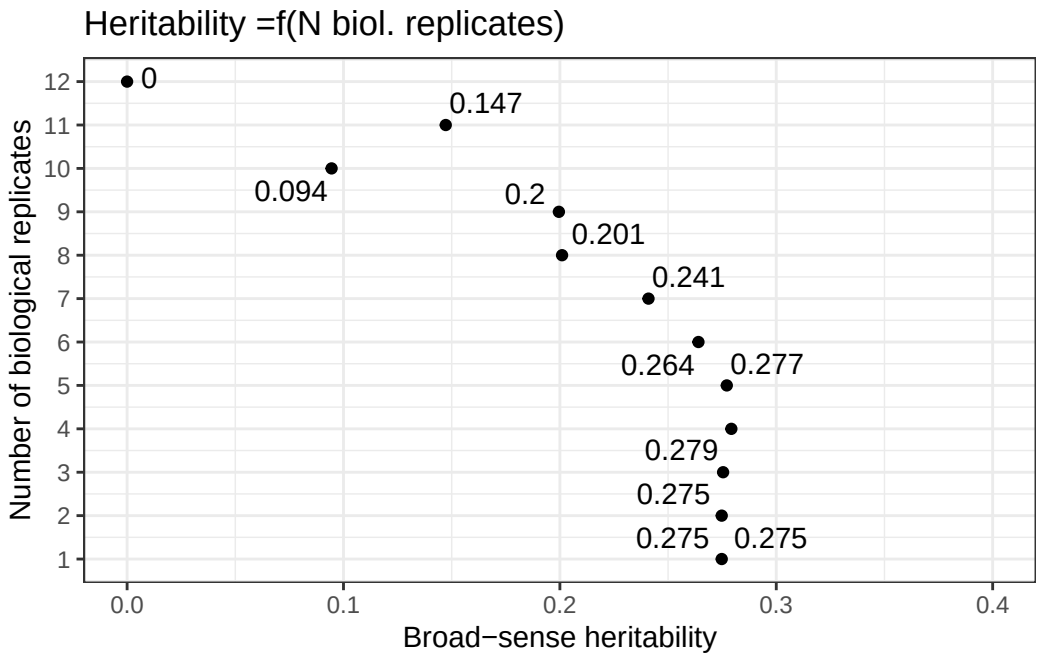
```
ggplot(dfb, aes(x = Nrep, y = Ngen)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,250)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = Ngen), arrow = T) +
```

```
  labs(x = "N biological replicates", y = "N genotypes") +
  ggtitle("N genotypes = f(N biological replicates)")
```



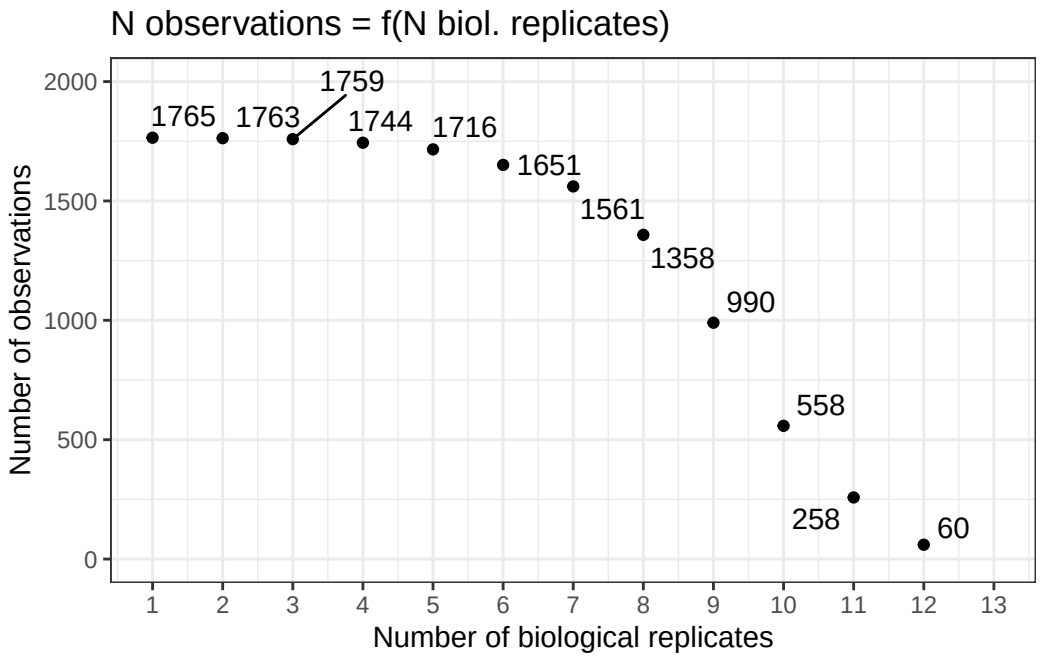
Heritability = f(N biological replicates)

```
ggplot(dfb, aes(x = H2, y = Nrep)) +
  geom_point() +
  scale_y_continuous(breaks=seq(1,13,1)) +
  scale_x_continuous(limits = c(0,0.4)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = round(H2,digits = 3))) +
  labs(y = "Number of biological replicates", x = "Broad-sense heritability") +
  ggtitle("Heritability =f(N biol. replicates)")
```



N observations = f(N biological replicates)

```
ggplot(dfb, aes(x = Nrep, y = Nob)) +
  geom_point() +
  scale_y_continuous(limits = c(0,2000)) +
  scale_x_continuous(limits = c(1,13), breaks = seq(from=1,to=13,by=1)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = Nob)) +
  labs(y = "Number of observations", x = "Number of biological replicates") +
  ggtitle("N observations = f(N biol. replicates)")
```



! Important

Based on these results, a threshold of >=5 biological replicates seems appropriate. That is keeping all genotypes for which at least 5 biological replicates are available.

Final filtering: keep genotypes with at least 5 repetitions

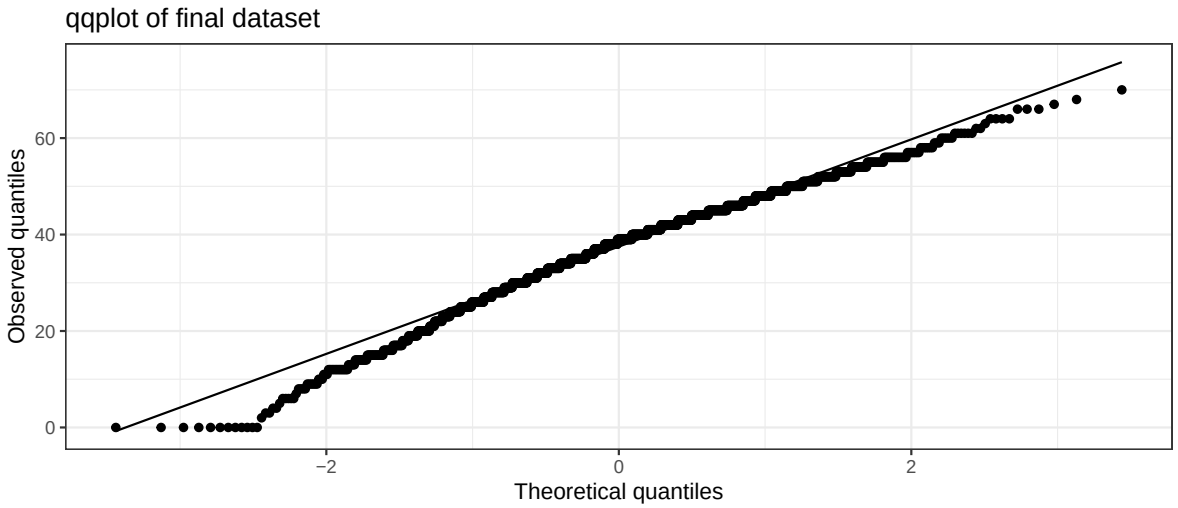
```
data250pisum_04 <- data250pisum_03 %>%
  group_by(genotype) %>%           # Group by genotype
  filter(n() >= 5) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250pisum_04$genotype <- droplevels(data250pisum_04$genotype)

res4 <- calculate_broad_heritability(data = data250pisum_04)
```

> **!** Important
>
> Conclusions:
> - Heritability is equal to 0.277.
> - There are 204 genotypes in total.
> - There are 1716 measurements in total.

```
create_correlation_plot(data250pisum_04,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
qqnorm(data250pisum_04$larvae_total_number)
```

```
p <- data250pisum_04 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("qqplot of final dataset") +
  theme_bw()
p
```


qqplot of final dataset
Observed quantiles
Theoretical quantiles
0
20
40
60
−2
0
2